

Liquid Biopsy Colorectal ctDNA Early Screening Report

Circulating Tumor DNA Methylation & Somatic Mutation Targeted Panel (Synthetic Demonstration)

Patient Demographics	Ordering Clinician	Specimen Chain of Custody
Name: Harold K. Sample DOB: 04/18/1964 (Age 62) Sex: Male Patient ID: SYN-MRN-5529184	Provider: Dr. Sophia Alvarez, MD Facility: Regional Cancer Center NPI: 9457896321 Location: Boston, MA	Specimen ID: SYN-SPEC-CRC-7731 Specimen Type: Blood / Plasma Tube: Cell-Free DNA BCT (10.0 mL) Collection Date: Jul 28, 2026 Received Date: Jul 29, 2026 Report Date: Aug 4, 2026

Result: Positive (Elevated Risk for Colorectal Malignancy)

Status: Final
Panel: Colorectal ctDNA Multi-Marker

Test Result Summary: Elevated circulating tumor DNA (ctDNA) markers detected in synthetic sample. Aberrant SEPT9 promoter hypermethylation and somatic mutations in KRAS and TP53 were identified in plasma cfDNA.

Quantitative and Qualitative Biomarker Findings

Target Marker	LOINC	Result	Reference Range	Interpretation
SEPT9 Methylation	77983-5	Positive	Negative	Abnormal (Positive)
KRAS Mutation Analysis	48018-6	p.G12D (VAF: 1.8%)	Not Detected (< 0.05% VAF)	Abnormal
BRAF Mutation Analysis	48018-6	Not Detected	Not Detected (< 0.05% VAF)	Normal
TP53 Mutation Analysis	48018-6	p.R273H (VAF: 2.3%)	Not Detected (< 0.05% VAF)	Abnormal
Total ctDNA Fraction	Local	2.1%	< 0.1%	Abnormal
Microsatellite Instability (MSI)	81695-9	MSI-Stable (MSS)	MSI-Stable	Normal
Tumor Mutational Burden (TMB)	Local	4.2 mut/Mb	< 10.0 mut/Mb	Normal

Clinical Interpretation & Recommended Next Steps

The presence of circulating tumor DNA with SEPT9 promoter methylation and concurrent KRAS p.G12D and TP53 p.R273H variants strongly indicates shed colorectal neoplastic DNA. Recommended next steps include:

- **Diagnostic colonoscopy:** High-definition mucosal inspection with targeted biopsy of any identified lesions.
- **Staging radiology:** Contrast-enhanced CT of abdomen and pelvis.
- **Therapeutic note:** The KRAS p.G12D mutation confers resistance to anti-EGFR antibody therapies (e.g. cetuximab, panitumumab).

Methodology & Limitations

Methodology: Plasma ctDNA undergoes dual-chemistry extraction. Methylation of Septin 9 is quantified using real-time PCR. Somatic variant profiling across hotspot exons of KRAS, BRAF, and TP53 is performed using deep amplicon NGS (minimum 25,000x coverage depth) on high-throughput sequencing systems.

Limitations: False-positive methylation or mutations may rarely occur due to clonal hematopoiesis of indeterminate potential (CHIP). Liquid biopsy results should be verified by endoscopic visualization and tissue histology.

Laboratory Director Authorization: Dr. Arthur Sterling, MD, FCAP | Electronically Signed | Date: Aug 4, 2026 | CLIA ID: 00D8874123